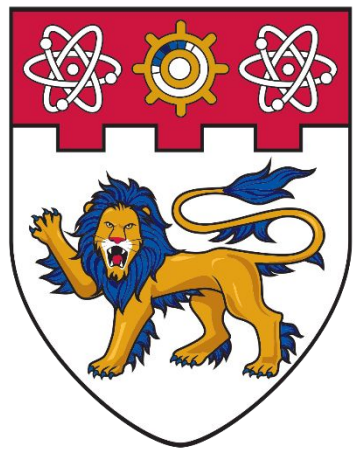




CN Yang Scholars Programme

Project title: Measuring Probabilistic Lives Saved from Risk Reduction Intervention in Cyclone Fani

Student name: Bernadeti Ausie Miranda Widawati

Supervisor: Prof David Lallemant

Mentor: Maricar Labalan Rabonza

Motivation

Assessing the effectiveness of Disaster Risk Reduction (DRR) interventions is essential to improve disaster outcome. However, the assessments mainly focus on monetary losses. This project aims to celebrate the success of the effective implementation of DRR efforts in reducing the loss of lives. Focusing attention on these successes will hopefully enable better appreciation of the ongoing DRR interventions.

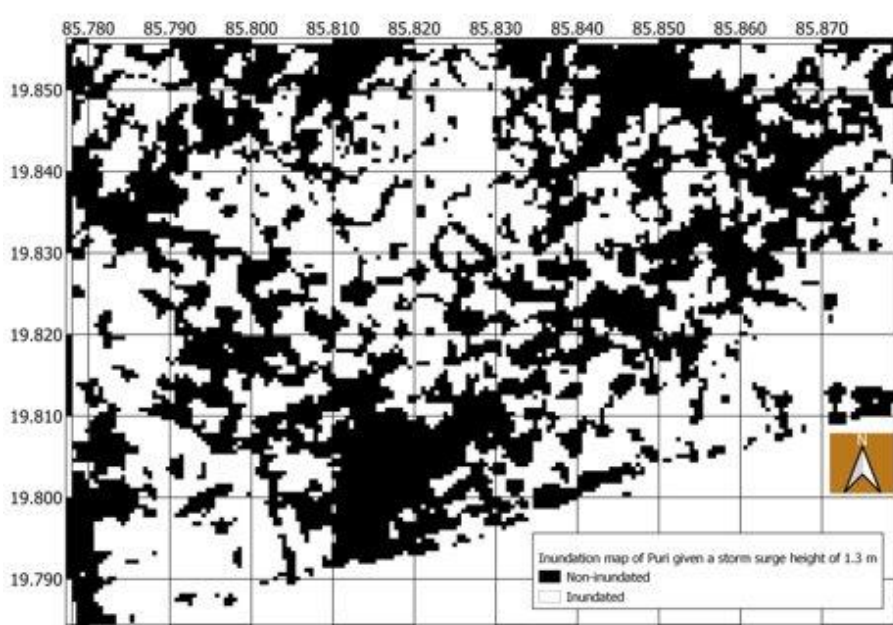
Objectives

The project focuses on studying the benefits of India's evacuation effort in Puri during the storm surge caused by Tropical Cyclone Fani which struck the state of Odisha on May 3, 2019. This project aims to establish a framework to estimate the mitigation effect of the evacuation process in terms of probabilistic lives saved.

Methodology

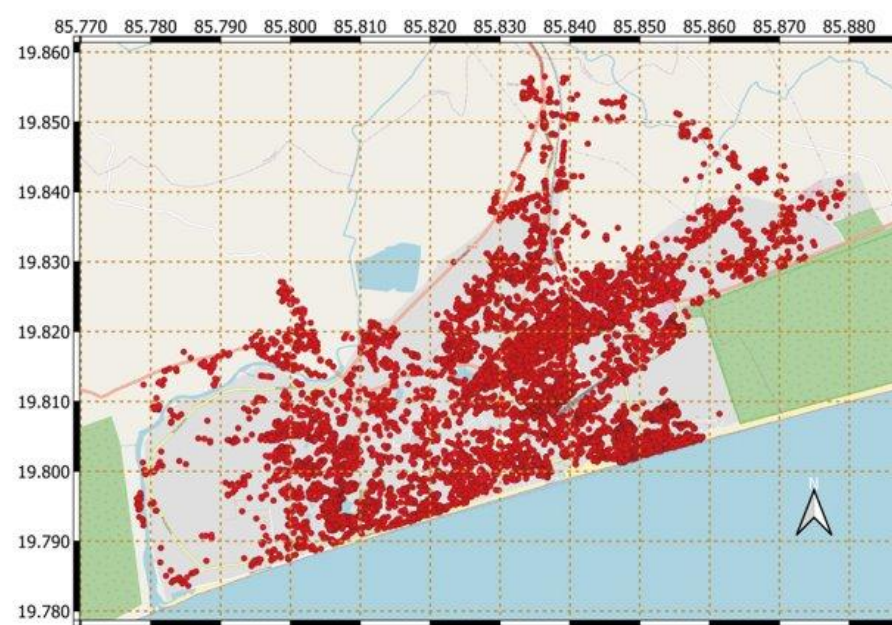
Counterfactual Disaster Risk Analysis

Freshwater drowning contributes to the most number of fatalities due to cyclone-related hazards. Collapsed buildings due to inundation might trap and injure their occupants. The methodology relates the storm surge inundation height and damage grading of inundated buildings to a fatality rate to obtain the number of probabilistic lives saved through evacuation



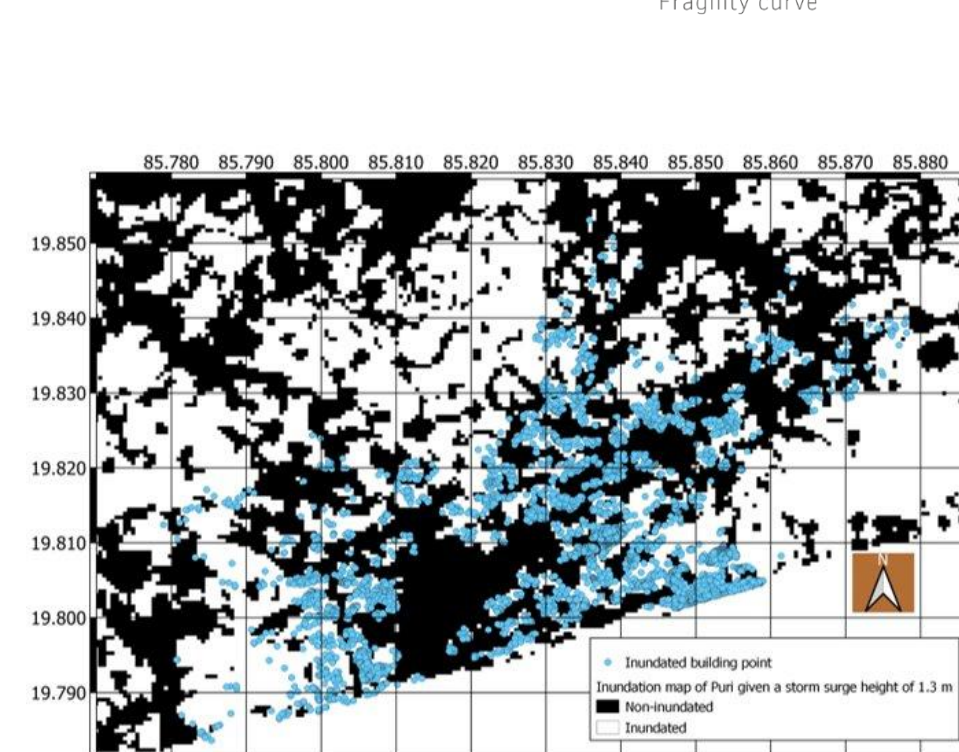
Hazard component

Fig 1. Modelled flood map of Puri developed using the GMRT DEM and recorded storm surge height of 1.3 m (GDACS)



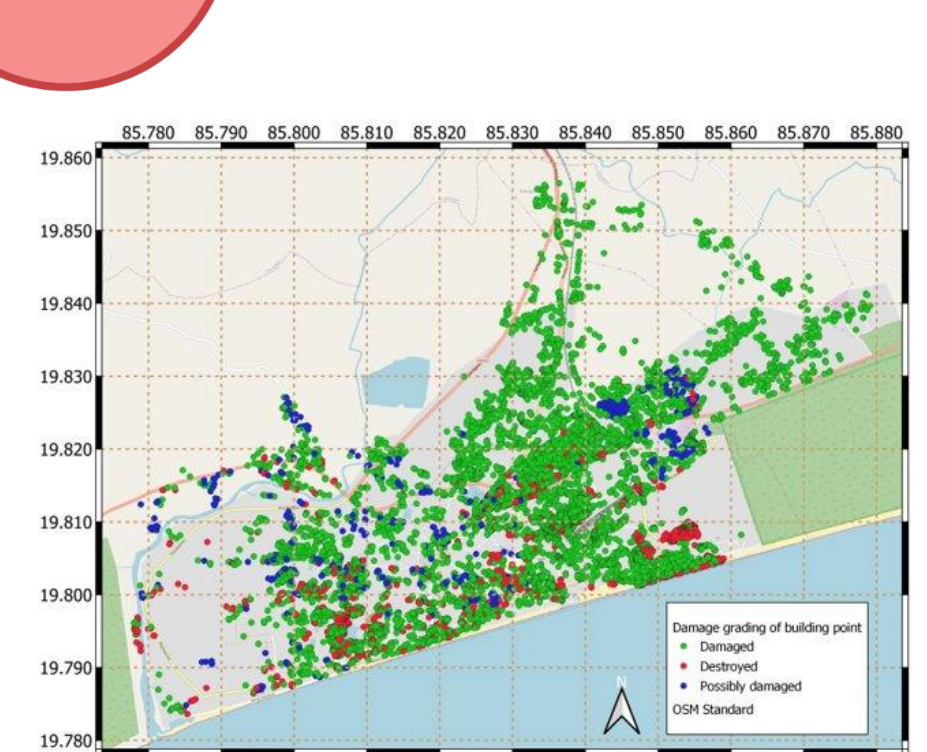
Exposure component

Fig 2. Building footprints of Puri, extracted from the Copernicus EMS Mapping. Occupancy rate of 4.5 people/building is used (IIPS)



Vulnerability component

Fig 3. Map of flooded buildings in Puri

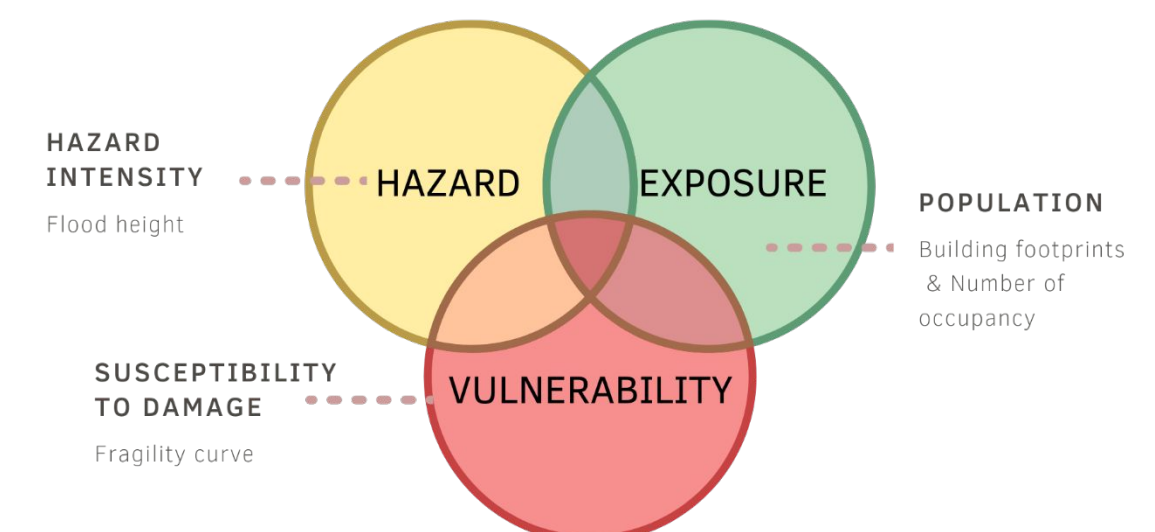


Vulnerability component

Fig 4. Spatial distribution of damage grading of buildings in Puri

Components of Disaster Risk Parameters

Risk = Hazard x Exposure x Vulnerability



Results & Analysis

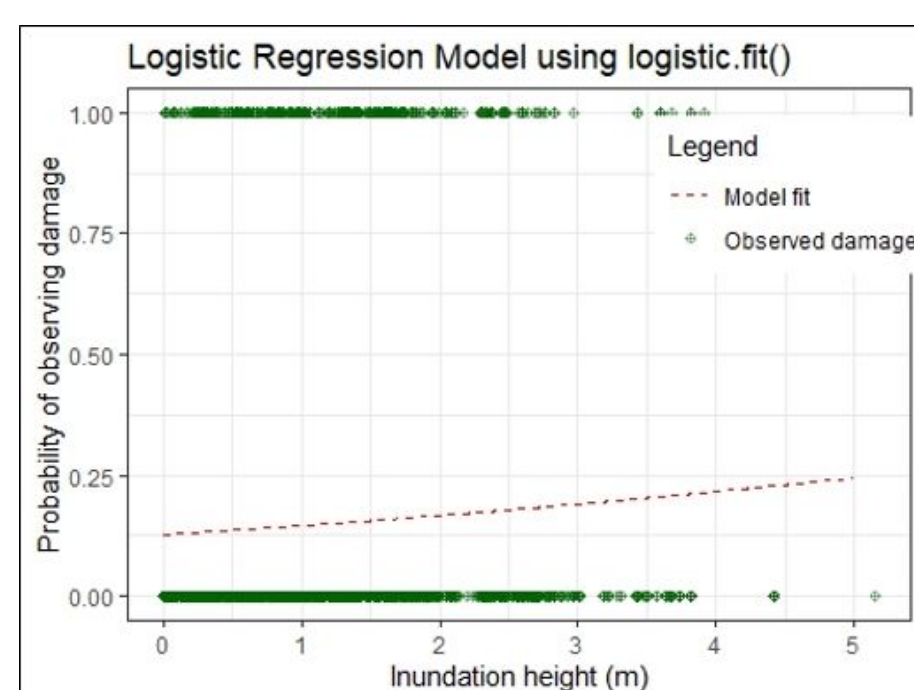


Fig 5. Fragility curve

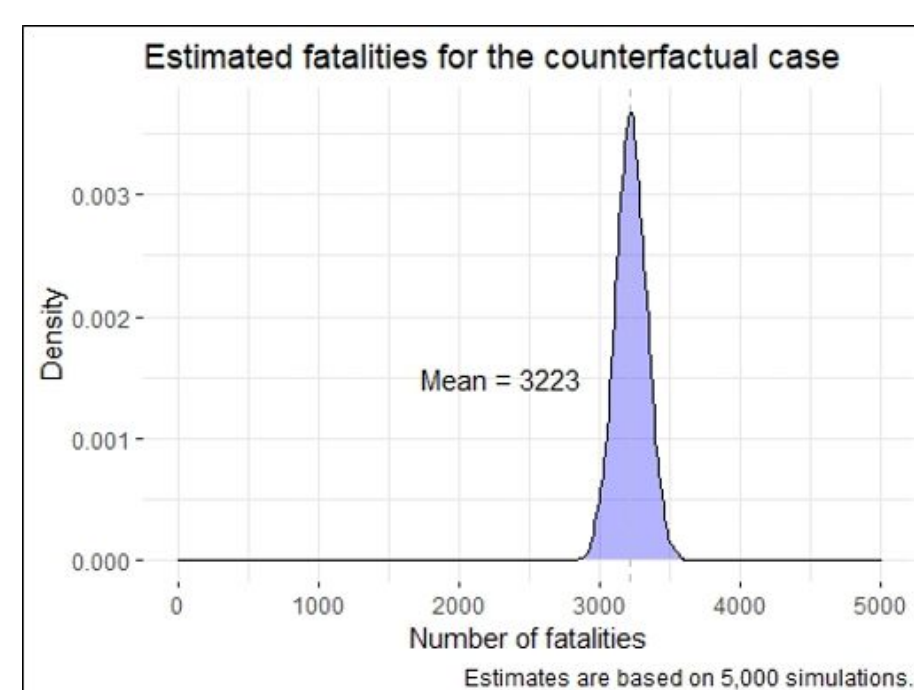


Fig 6. Probability density graph of estimated counterfactual fatalities

$$P_{\text{observing damage}} = \frac{1}{1 + e^{-(-1.9269206 + 0.1603181 \cdot \text{inundation height})}}$$

- Mean estimated fatalities for the counterfactual case of no evacuation: 3223
- Actual fatalities: 39
- Probabilistic lives saved due to evacuation: 3223-39 = 318420

Conclusion

A counterfactual disaster risk analysis is adopted in the study of the impacts of evacuation in saving lives during a storm surge flooding caused by Cyclone Fani. We have established a framework of counterfactual disaster risk analysis to quantify the benefits of DRR interventions in terms of lives saved, hence enabling the celebration of their successes. Through statistical analysis of the modelled inundation height (hazard component of risk) and the damage grading of each building point of certain occupancy rate (exposure component of risk), a fragility curve relating inundation height and the probability of observing damage (Fig 5) and a probability density graph of the counterfactual number of fatalities (Fig 6) in Puri are generated.

Future Work

More accurate modelling of storm surge inundation height and a better-fitted fragility curve can be developed. We can apply Puri's fragility curve to measure probabilistic lives saved in other districts in India.